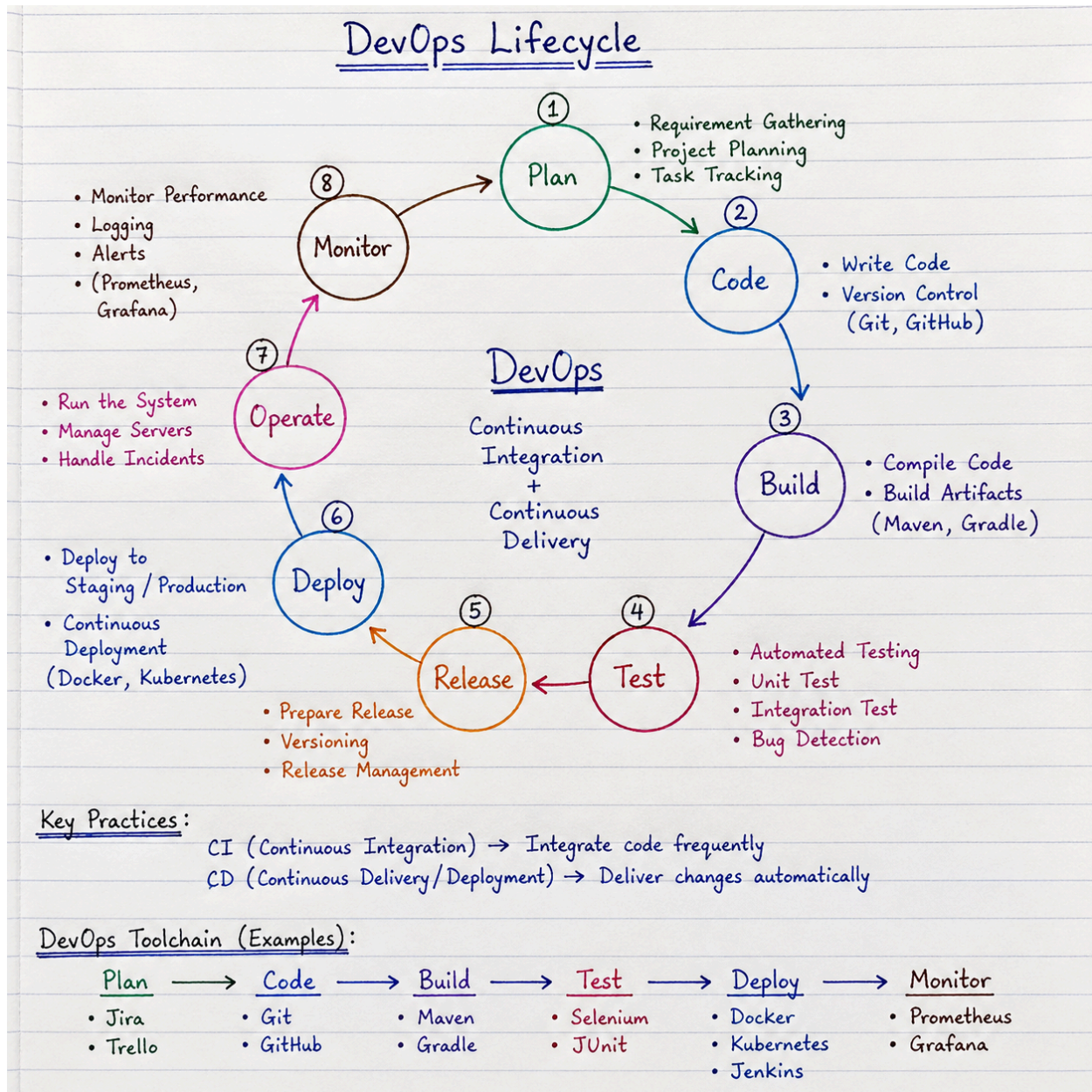


DevOps



DevOps (Development + Operations) is a software development approach that:

- Integrates **development and IT operations teams**
- Focuses on **automation, collaboration, and continuous delivery**

Goal: Deliver software **faster, reliably, and efficiently**

Continuous Integration (CI)?

CI is a practice where:

- **Developers frequently merge code into a shared repository**
- **Each change is automatically tested**

Key Points:

- **Early bug detection**
- **Automated builds & tests**
- **Improves code quality**

Example tools: Jenkins, GitHub Actions

Continuous Delivery (CD)?

CD is a practice where:

- **Code changes are automatically prepared for release**
- **Software is always in a deployable state**

Key Points:

- **Automated deployment pipeline**
- **Faster and reliable releases**
- **Minimal manual intervention**

Example tools: Docker, Kubernetes